BEE 4750/5750

Homework 3: Discretization, Numerical Convergence, and Simulation

2026-09-17

Due Date
Thursday, 09/24/26, 9:00pm

Overview

Instructions

- Problem 1 asks you to discretize several systems models by hand and reason about characteristic timescales and convergence, without writing any code.
- Problem 2 asks you to extend the shallow lake model from class to include a sediment phosphorus stock, then use numerical simulation to check convergence and study the resulting long-term lake dynamics.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
using Plots
using LaTeXStrings
```

Problems (Total: 50 Points)

Problem 1 (15)

For each of the following systems models, (i) write the forward Euler update rule, (ii) find the (local) characteristic timescale and use it to evaluate whether the given step size Δt is a reasonable choice, and (iii) use the given error to predict the error at the new step size, assuming the method is well into its first-order convergence regime.

Problem 1.1 (4)

$$\frac{dC}{dt} = 5 - 0.25C$$

(a) Write the forward Euler update rule for $C(t + \Delta t)$.

(b) Find the characteristic timescale of this system. Is $\Delta t = 1$ a reasonable starting choice of step size? Justify your answer.

(c) A forward Euler simulation of this system has an error of 0.6 at $\Delta t = 2$. What error would you expect at $\Delta t = 1$?

Problem 1.2 (5)

A well-mixed holding tank has volume $V = 200\ \mathrm{m}^3$ and a through-flow of $Q = 50\ \mathrm{m}^3/\ \mathrm{yr}$ carrying an inflow concentration $C_{\mathrm{in}} = 2\ \mathrm{mg/L}$. A pollutant in the tank also decays at a first-order rate $k = 0.3\ \mathrm{yr}^{-1}$. The concentration in the tank, $C(t)$, is governed by

$$\frac{dC}{dt} = \frac{Q}{V}(C_{\mathrm{in}} - C) - kC.$$

(a) Write the forward Euler update rule for $C(t + \Delta t)$.

(b) Find the characteristic timescale of this system. Is $\Delta t = 1\ \mathrm{yr}$ a reasonable starting choice of step size? Justify your answer.

(c) A forward Euler simulation of this system has an error of 0.8 at $\Delta t = 1.6\ \mathrm{yr}$. What error would you expect at $\Delta t = 0.4\ \mathrm{yr}$?

Problem 1.3 (6)

$$\frac{dx}{dt} = 3 - 0.5x^2, \qquad x \geq 0$$

(a) Write the forward Euler update rule for $x(t + \Delta t)$.

(b) This system's rate of change depends on the state x, so there is no single characteristic timescale for the whole system — only a *local* one. Find the system's equilibrium (be careful about the problem domain), then linearize near it to find the local feedback gain and the local characteristic timescale there. Is $\Delta t = 0.5$ a reasonable starting choice of step size near this equilibrium? Justify your answer.

(c) A forward Euler simulation of this system has an error of 1.5 at $\Delta t = 0.3$. What error would you expect at $\Delta t = 0.1$?

Problem 2 (35)

In class, we modeled a shallow lake's phosphorus (P) concentration with

$$\frac{dP}{dt} = L - sP + R(P),$$

treating the maximum sediment release rate r (inside $R(P)$) as a fixed constant. A more advanced version of this model could be built by noting that r is really κM, where M is a **sediment phosphorus pool** — a slowly-filling stock.

The two-stock model splits the single loss rate s into two pieces (outflow/permanent burial, s_{out}, and transfer into the sediment, h), and adds a second state variable $M(t)$ for the sediment pool:

$$\frac{dP}{dt} = L - s_{\mathrm{out}}P - hP + \kappa M\,\frac{P^q}{m^q + P^q}$$
$$\frac{dM}{dt} = hP - bM - \kappa M\,\frac{P^q}{m^q + P^q}$$

where b is the rate at which sediment P is permanently buried. Use the parameter values in Table 1 for the rest of this problem.

Table 1: Parameters for the two-stock lake model.

Parameter	Meaning	Value
s_{out}	P outflow/permanent burial rate (from the water column)	$0.3\ \text{yr}^{-1}$
h	P transfer rate into the sediment	$0.3\ \text{yr}^{-1}$
b	sediment P burial rate	$0.02\ \text{yr}^{-1}$
κ	sediment recycling rate coefficient	$0.15\ \text{yr}^{-1}$
m	P concentration at half-maximum sediment release	$25\ \mu\ \text{g}\ /\ \text{L}$
q	steepness of the sediment release response	8

Assume the lake starts with no phosphorus anywhere in the system, $P(0) = M(0) = 0$.

Problem 2.1 (5)

Derive the forward Euler update rules for $P(t + \Delta t)$ and $M(t + \Delta t)$. Note that these two updates are coupled: at each step, both updates use the *current* $P(t)$ and $M(t)$ (not the newly-updated value of one when computing the other).

Problem 2.2 (13)

Implement your update rules from Problem 2.1 and use them to simulate the lake under a loading of $L = 10\ \mu\ \text{g}\ /(\text{L} \cdot \text{yr})$, starting from $P(0) = M(0) = 0$.

Two different quantities have to be "large enough" here, for two different reasons:

- the **simulation horizon** T has to be long enough for the slow sediment stock M to fill. Its characteristic timescale is $1/b = 1/0.02 = 50\ \text{yr}$, so use $T = 300\ \text{yr}$ — six of those timescales. You will reuse this same run in Problem 2.3.
- the **step size** Δt is what you are actually checking for numerical convergence, at that fixed T.

Run a convergence study on $P(30)$, the phosphorus concentration thirty years in. Build a reference solution at a very fine step size (smaller than you think you might need), then test a sequence of successively-halved, coarser step sizes against it. Here is a starting point:

```
function simulate_lake(loading, horizon, step)
    # TODO: forward Euler on P and M, using your update rules from Problem 2.1.
    # Both updates use the *current* P and M, not a partly-updated value.
    # Return the full P trajectory as a vector, one entry per step taken.
    error("simulate_lake has not been written yet -- replace this line with your loop")
end

step_reference = 0.001                          # 300,000 steps at T = 300; a few seconds
P_reference = simulate_lake(10.0, 300.0, step_reference)
P30_reference = P_reference[Int(30 / step_reference)]     # the entry at t = 30 yr
```

Report a table of your step sizes, the resulting $P(30)$, the error relative to your reference, and the empirical order of convergence between successive refinements. Make a log-log plot of error vs. Δt, and state which step size you will use for the rest of this problem and why.

> ⚠ A step size that is too large will not just be inaccurate
>
> Forward Euler can blow up entirely, rather than merely losing accuracy, when the step is too large for the *fastest* process in the system. Here that is P itself, with a timescale of $1/(s_{\text{out}} + h) \approx 1.7\ \text{yr}$ — much faster than the sediment pool that set T. **The horizon is governed by the slowest process; the step size is governed by the fastest one.**
>
> Start your sequence at $\Delta t = 2\ \text{yr}$ and halve from there. If you try something much coarser you may see the simulation diverge or oscillate wildly; that is worth a sentence, but it is not what the convergence study is measuring.

Problem 2.3 (10)

Using the step size you justified in Problem 2.2, plot $P(t)$ over the full 300-year simulation.

You already have $P(30)$ from Problem 2.2. What does P look like once the simulation has fully run? What is happening physically to the lake between those two points in time, and around when does it happen?

Problem 2.4 (7)

Suppose a lake manager instead used a simpler one-stock model that ignores the sediment pool entirely (as if $\kappa = 0$, so there is no recycling at all):

$$\frac{dP}{dt} = L - s_{\text{out}} P - hP.$$

What is this simplified model's equilibrium P at $L = 10\ \mu\ \text{g}\ /(\text{L} \cdot \text{yr})$, and would the manager have concluded the lake is safely oligotrophic at this loading (recall that sediment release becomes significant once P approaches $m = 25\ \mu\ \text{g}\ /\ \text{L}$)?

Compare this to what you found in Problem 2.3. What does this tell you about the risk of using a simplified model, or a short simulation, to set a "safe" loading level? What would you recommend a lake manager do differently, given what you now know about this system?

References

List any external references consulted, including classmates.